ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Fourier Filtering

Image Processing & Analysis for Life Scientists

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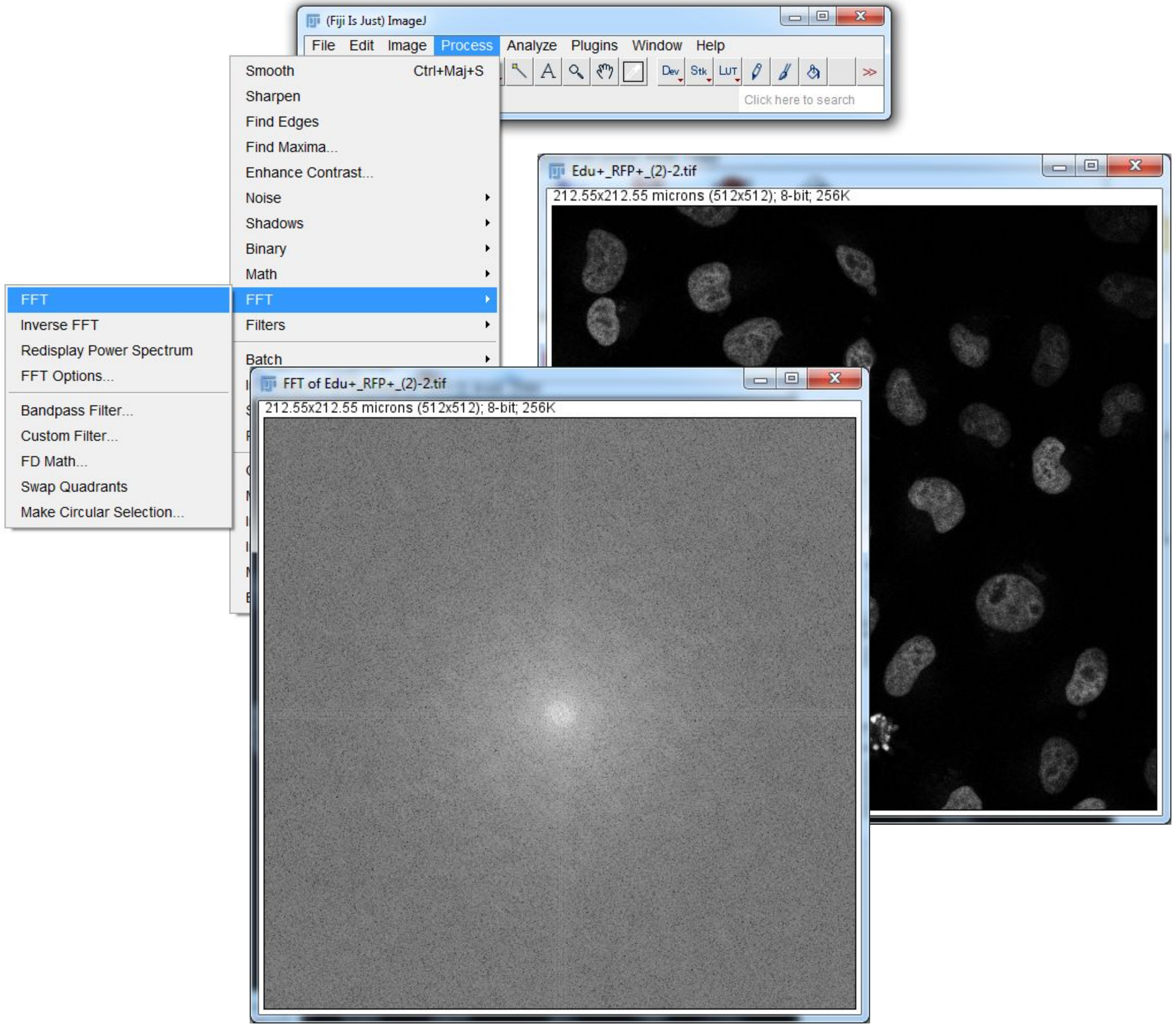
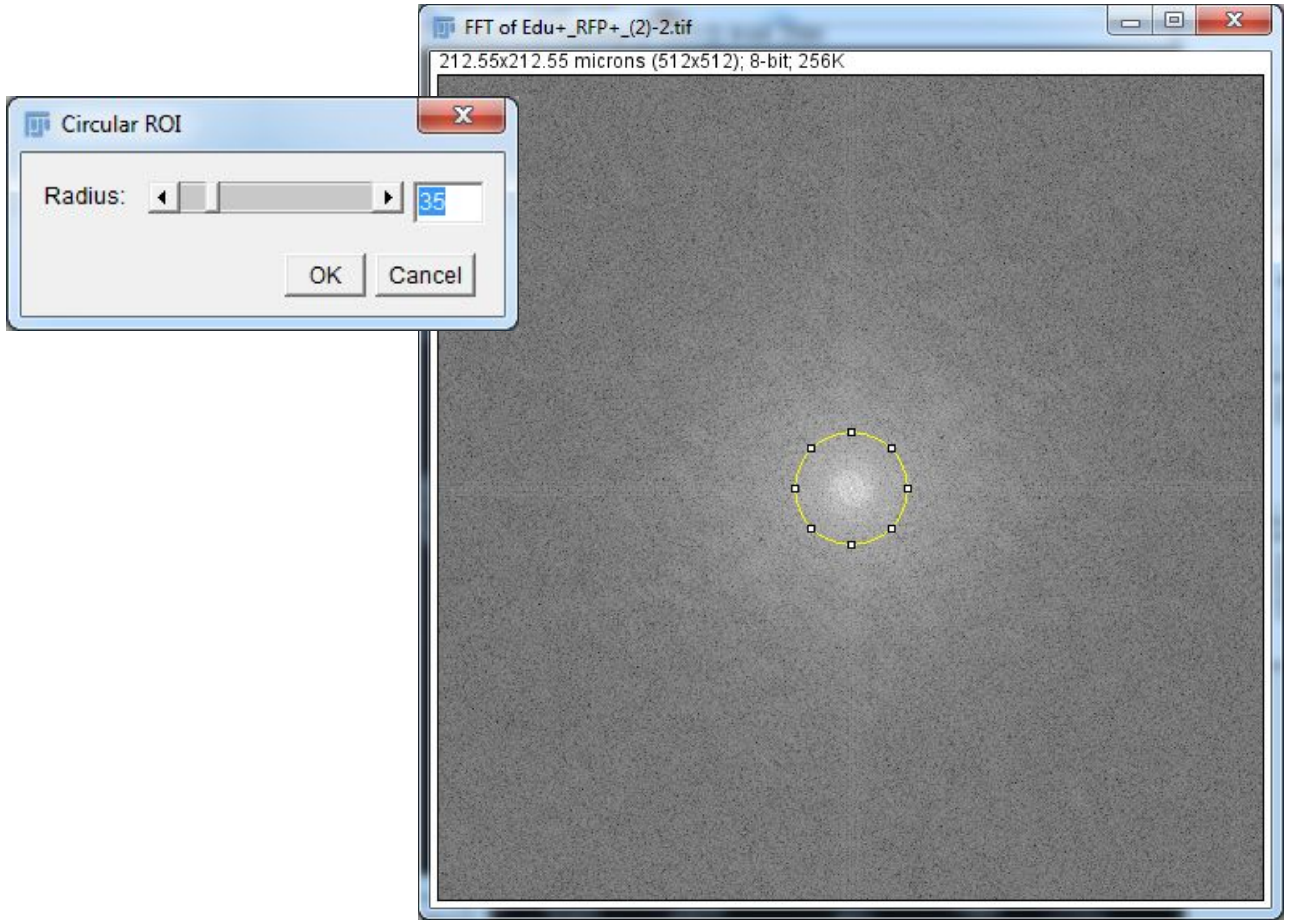
Getting the FFT

1. Open **an image from HeLa_EdU**
   Duplicate the first channel only
2. Go to Process > FFT > FFT
3. Observe the FFT image



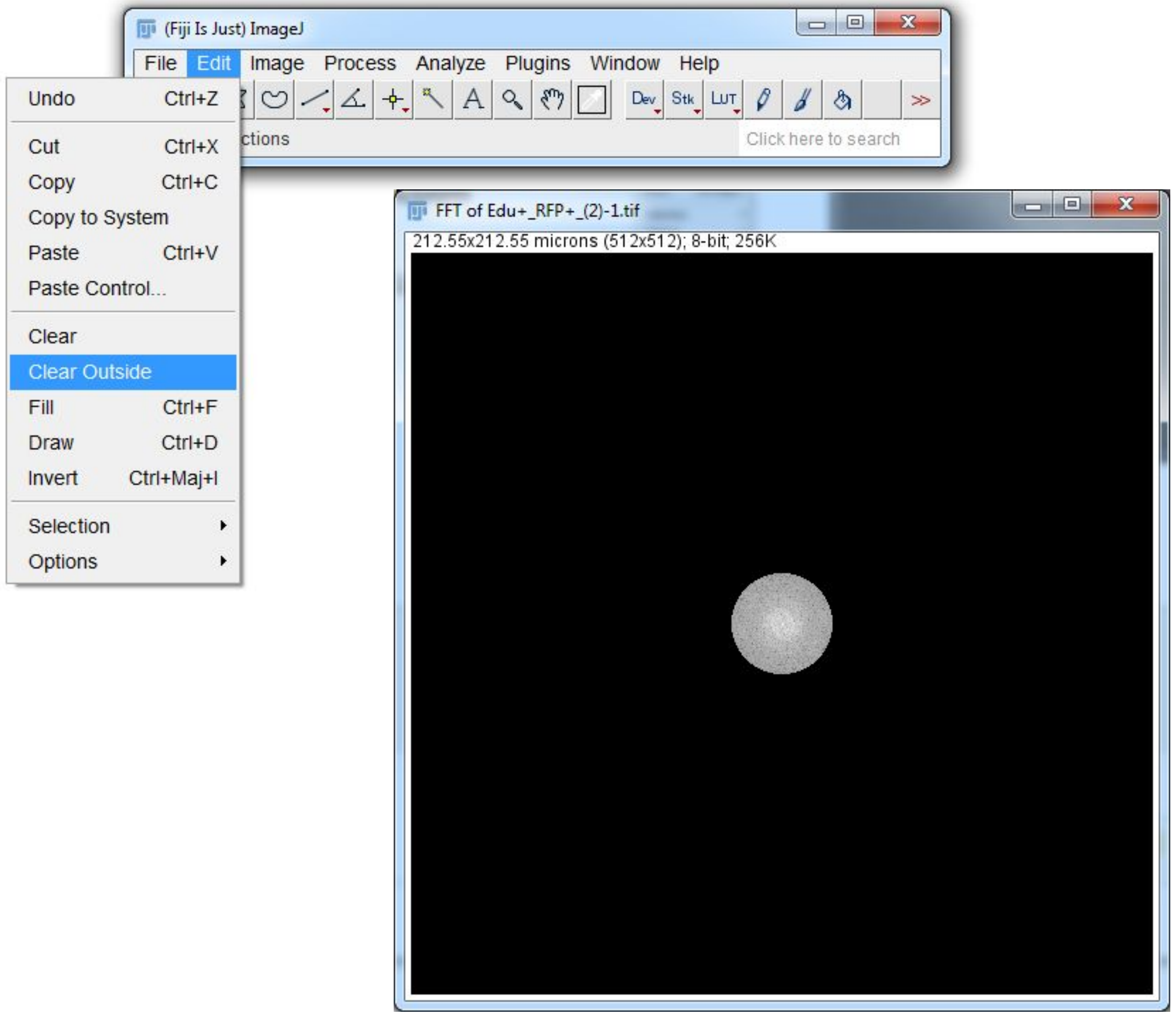
Modifying the FFT

1. Open **an image from HeLa_EdU**
   Duplicate the first channel only
2. Go to **Process > FFT > FFT**
3. Observe the FFT image
4. Make a circle using
   Process>FFT>Make Circular Selection



Modifying the FFT

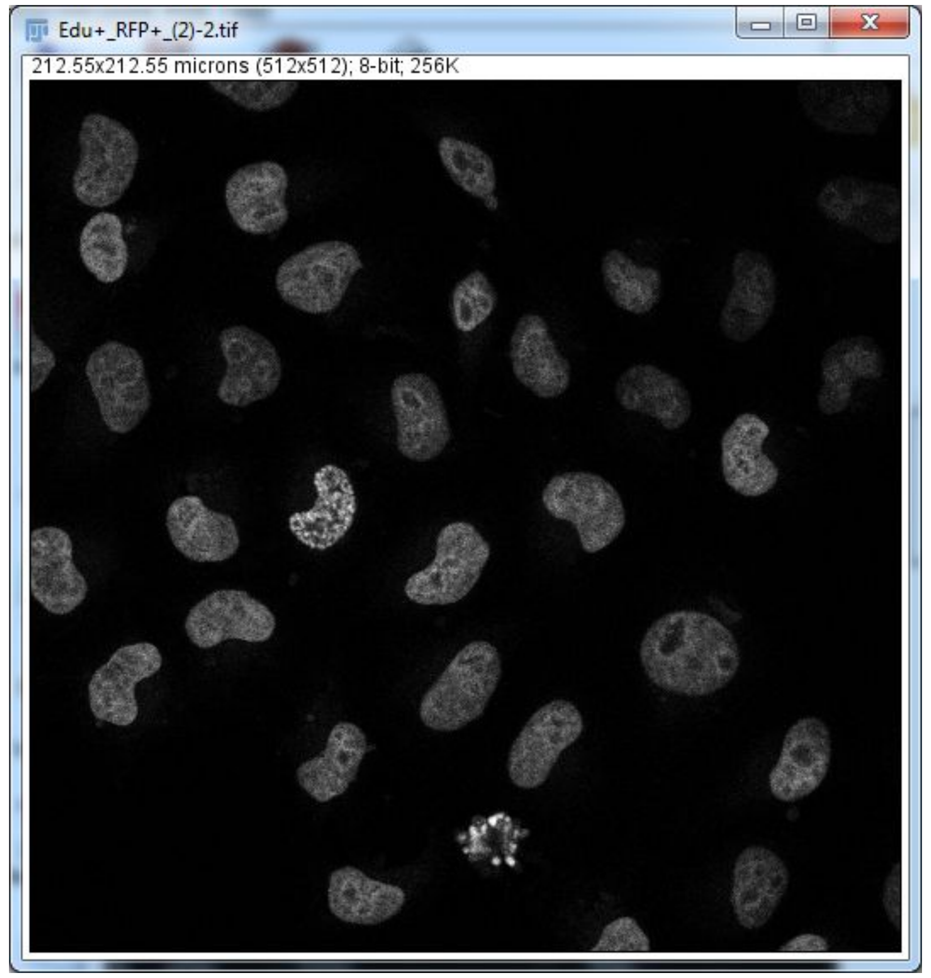
1. Open **an image from HeLa_EdU**
   Duplicate the first channel only
2. Go to **Process > FFT > FFT**
3. Observe the FFT image
4. Make a circle using
   Process>FFT>Make Circular Selection
5. Use **Edit>Selection>Clear Outside**
   Make sure the background is black using the color picker



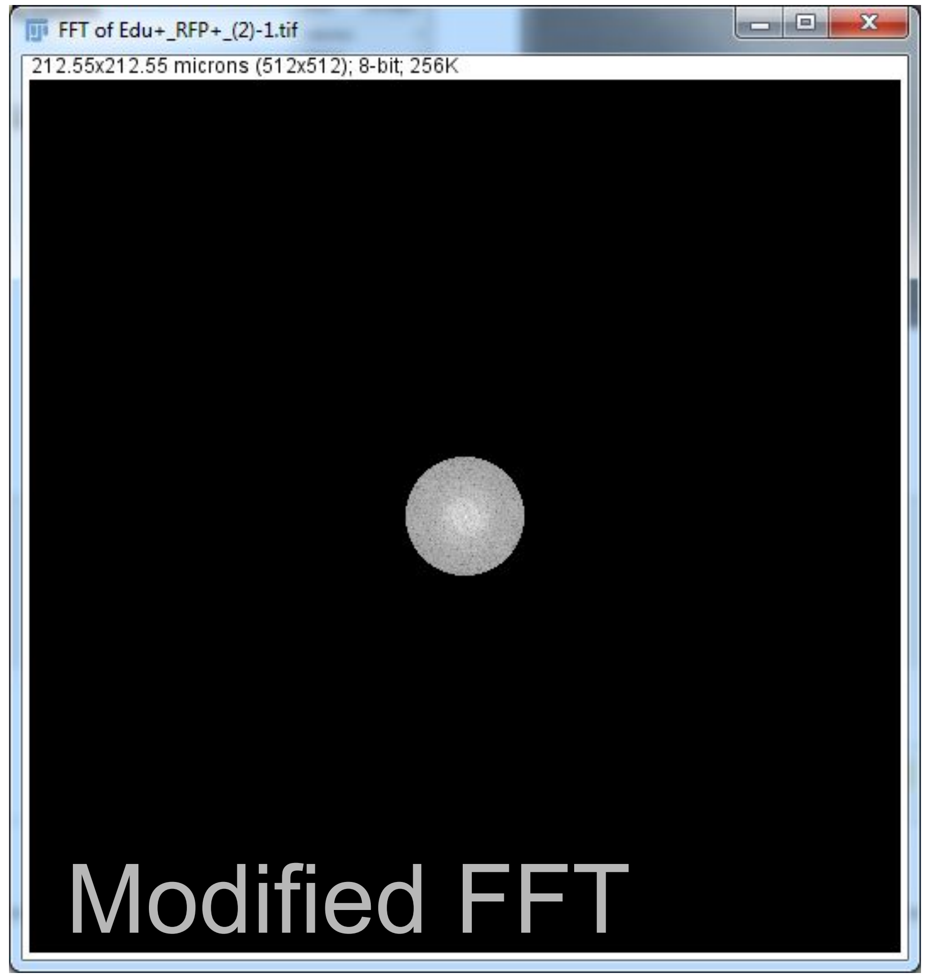
The Inverse FFT

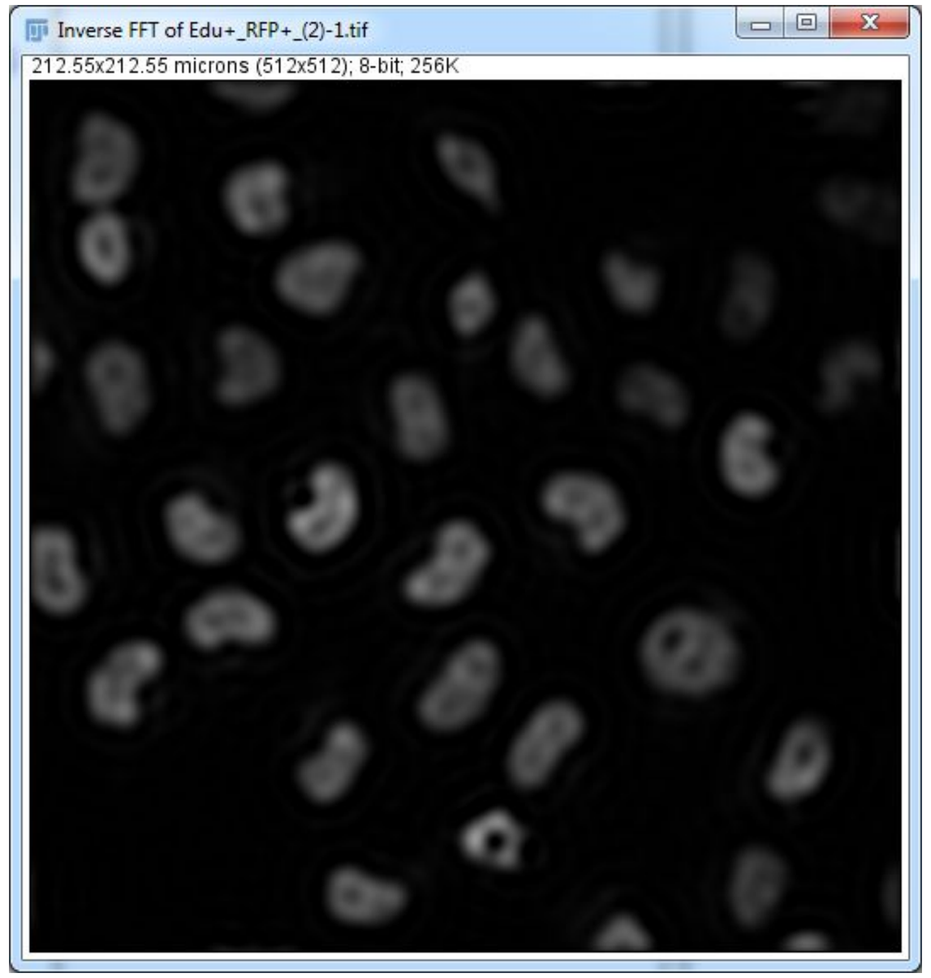
1. Open **an image from HeLa_EdU**
   Duplicate the first channel only
2. Go to **Process > FFT > FFT**
3. Observe the FFT image
4. Make a circle using
   Process>FFT>Make Circular Selection
5. Use **Edit>Selection>Clear Outside**
   Make sure the background is black using the color picker
6. Use **Process>FFT>Inverse FFT**

We have applied a **Lowpass** filter: All high frequency components have been removed



Original

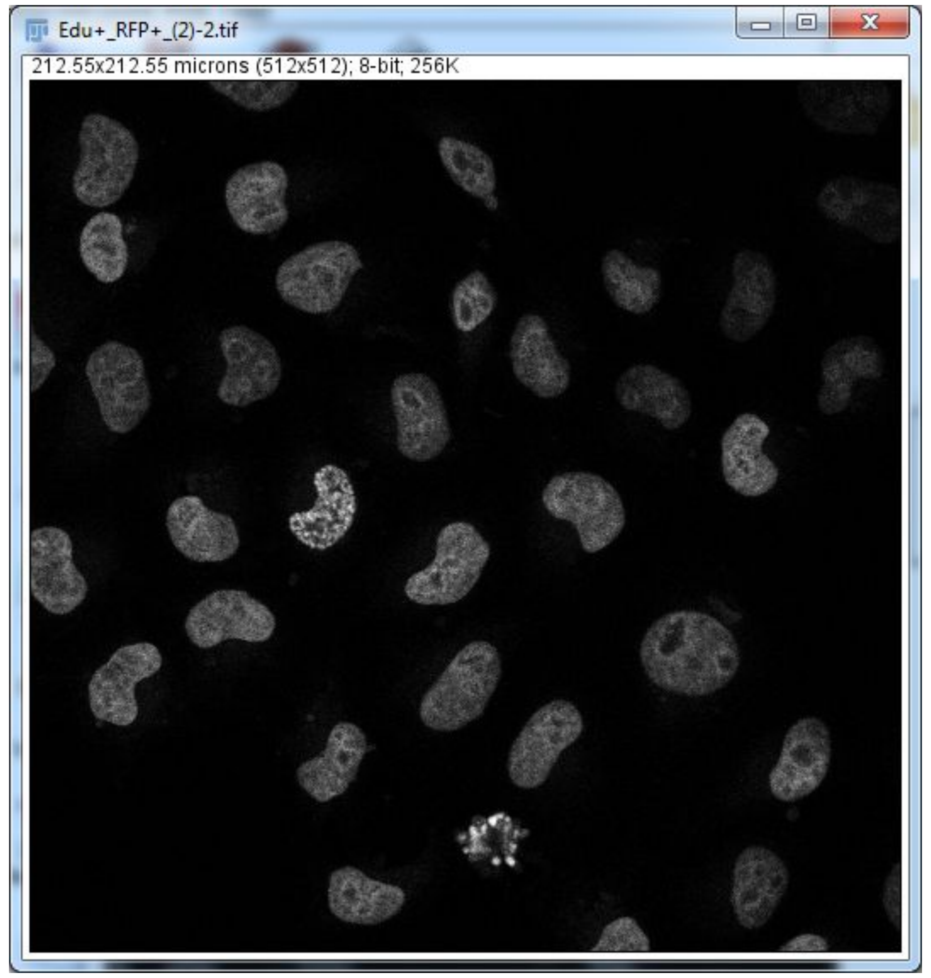




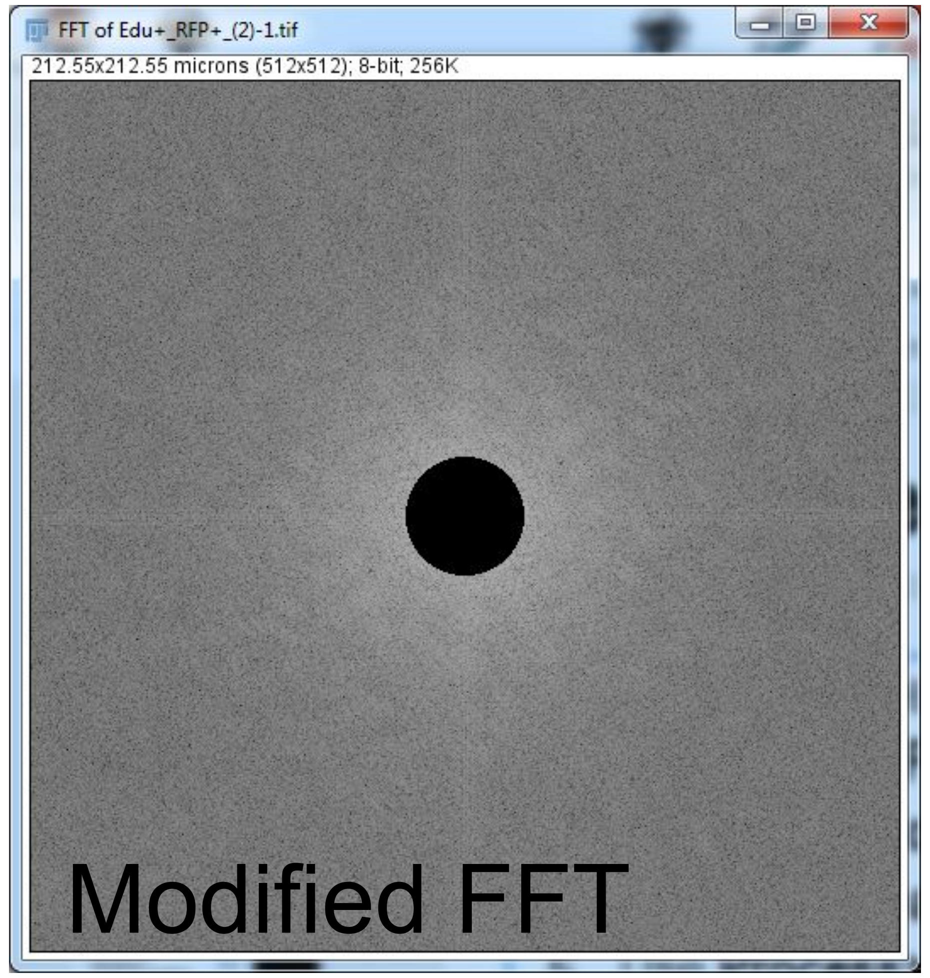
Filtered

The Inverse FFT

1. Go back to the original image
2. **Process > FFT > FFT**
3. **Edit >Selection>Restore Selection**
4. Use **Edit>Clear**
5. Use **Process>FFT>Inverse FFT**



Original

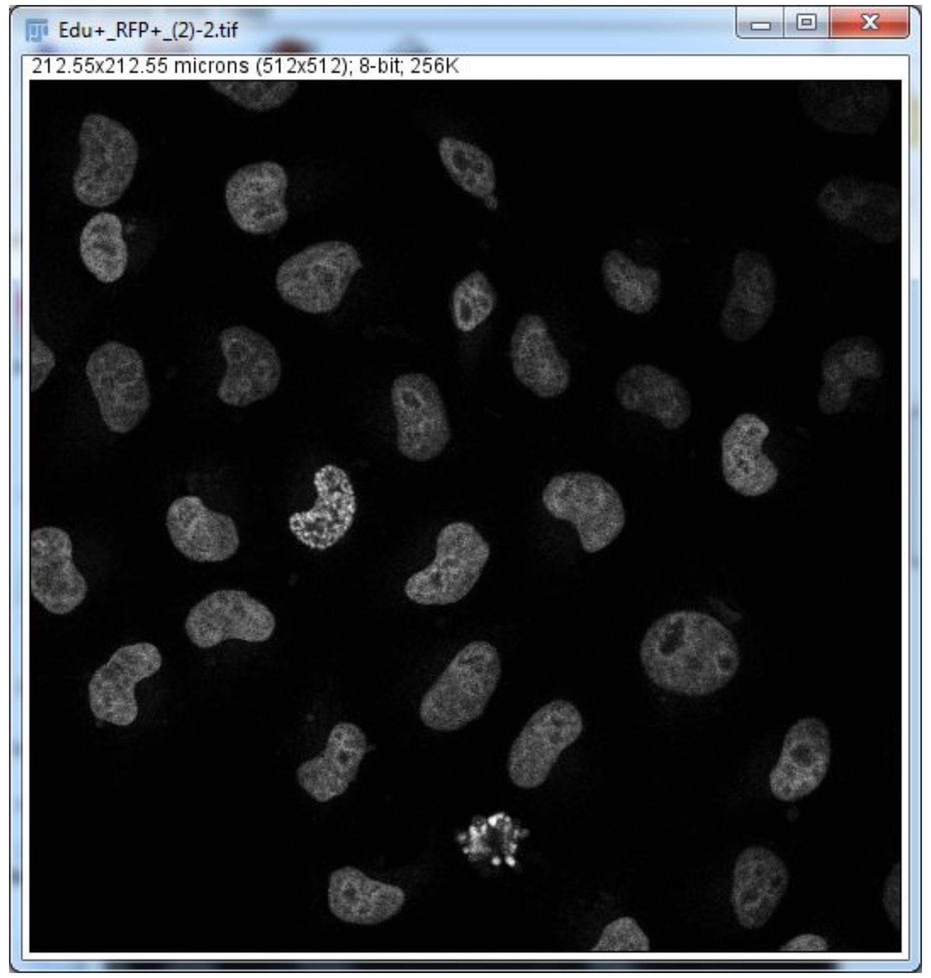


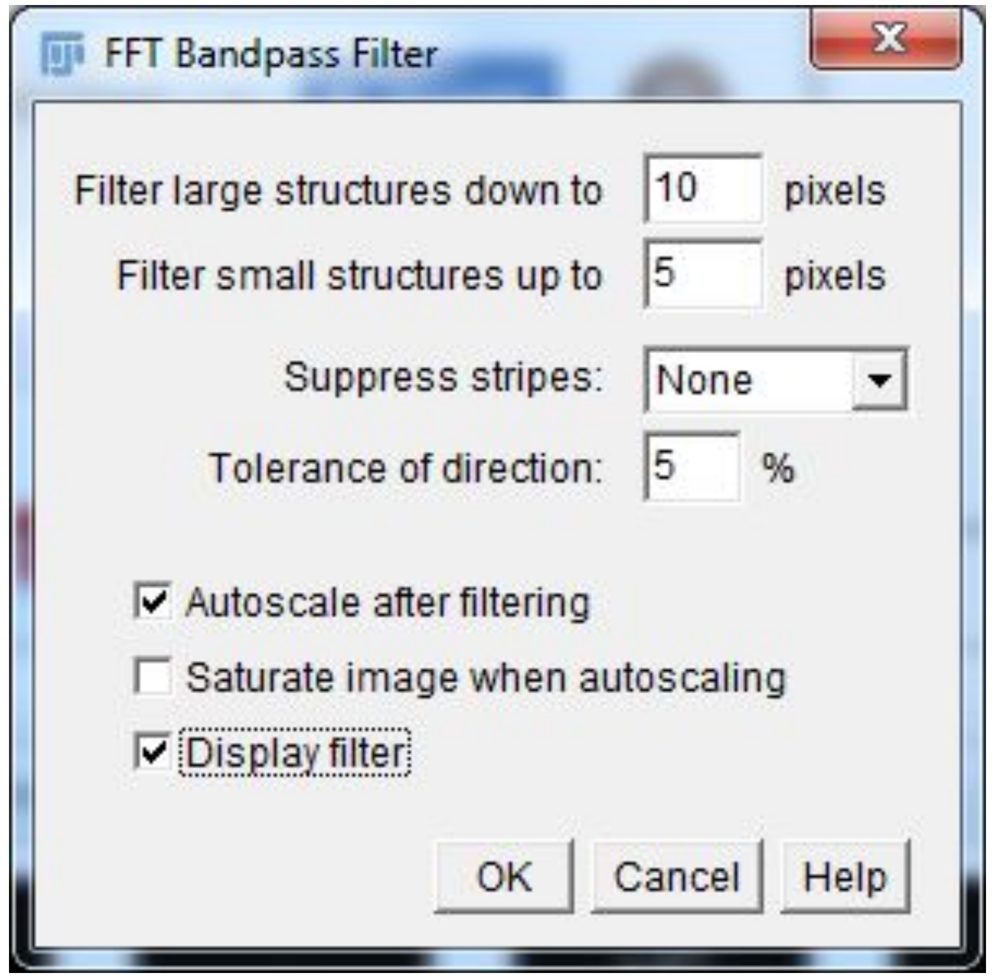
Filtered

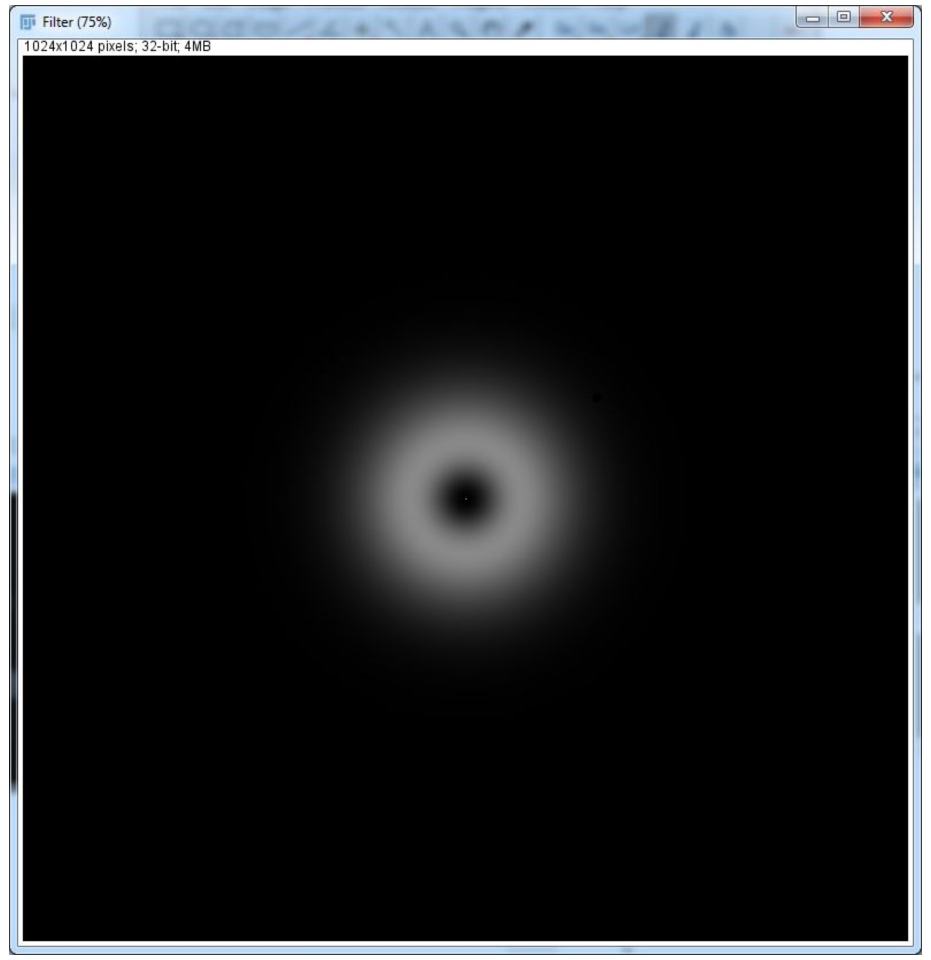
We have applied a **Highpass** filter: All low frequency components have been removed

The FFT Bandpass Filter

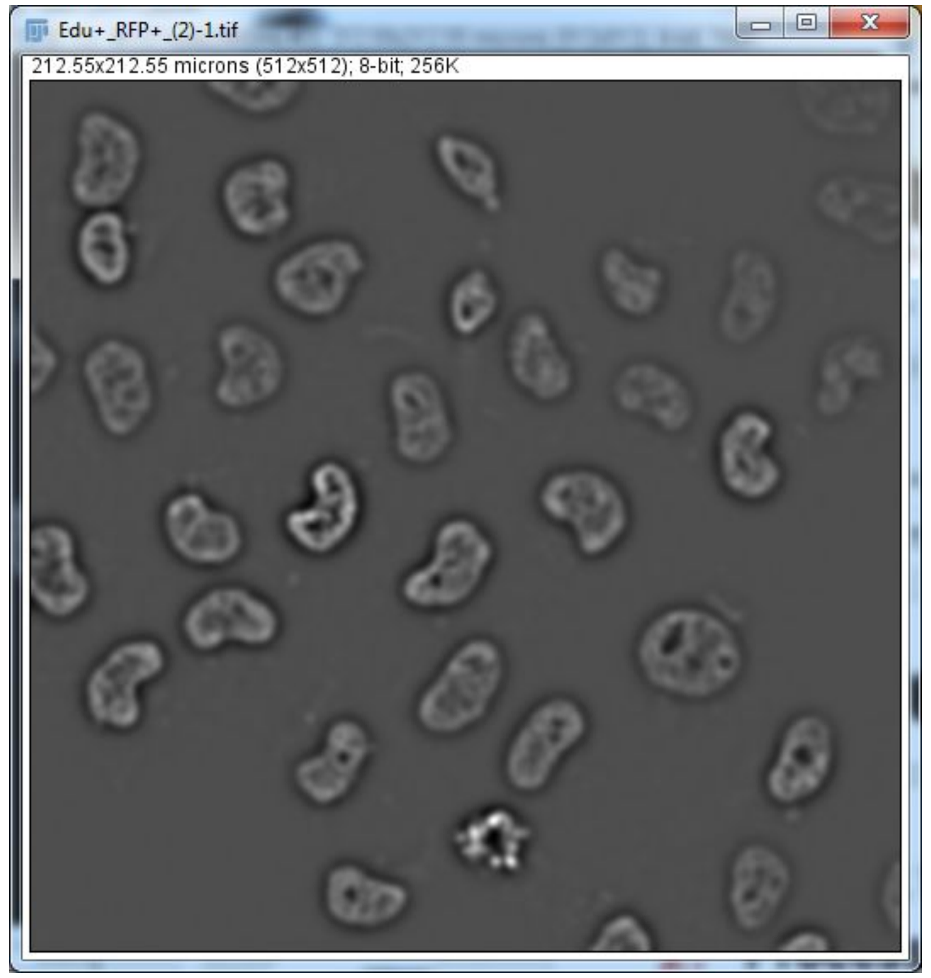
1. Go back to the original image
2. **Process > FFT > Bandpass Filter**
3. **Use settings as shown**
4. Click OK
5. Use **Process>FFT>Inverse FFT**







Resulting Filter



Resulting Image

We have applied a **Bandpass** filter: All periods between 5 and 10 pixels have been kept.